University of Kufa
Faculty of Computer Science and Mathematics
Department : Computer Science
Course Name : Software Engineering

Academic Year : 2021 / 2022
Course Stage / Level : 1st
Date of Exam : 15 / 5 / 2022
Time : 1 Hours

Note: Answer all questions.

Q1: Choose between brackets (5 marks)

1. If you create an array A= new int [3] you should think of it as a "*Array*" with----------

(a.3 rows and 1 column b. 1 row 3 columns c. 3 rows 3 columns d. 1 row 1 column)

2. To removes any spaces at the beginning or end of a string using --------- method.

a.trim() b. equals() c. endsWith() d.startsWith())

3. charAt(i) Returns the character at that position (i) in the string as -------.

(a. int b.string c. char d. boolean)

4. The action of changing a data type into a different type is called-------.

(a. type Conversion b. break c. continue d. type casting)

5. The output of the following statement System.out.println(Math.ceil(50.7)); is----------.

(a. 50 b. 51 c. 51.0 d. 50.0)

Q2 : Write down a Java program to read a 2-dimensional array x[4][4] then perform each of the following:(15 marks)

1. Find the maximum and minimum in the third row.
2. print positive numbers only in the main diagonal. فقط رئيسي موجب
3. Replace the upper triangle elements with 1.

-1	3	5	6
5	-2	4	6
3	4	5	7
1	7	9	-3

Q3. Find the output of the following code: (10 marks)

A)

```
for(i=1; i<=5; i++)
  { for(j=1; j<=i; j++)
      System.out.print(i+j);
      System.out.println();
  }
```

1 2 3 4 5
2
3 4
4 5 6
5 6 7 8
6 7 8 9 10

B)

```
int[] Array = {2, 5, 1, 9, -11};
System.out.println(Array.length);
System.out.println(Array[-2]);
```

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استاذ المادة
م.م. هدى عبد الحسين

رئيس القسم
أ.م.د. اسعد نوري هاشم

University of Kufa
Faculty of Computer Science and Mathematics
Department : Computer Science
Course Name: Advanced programming language

Academic Year : 2021 / 2022
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Note: Answer all questions.

Q1 TRUE OR FALSE (5 marks)

1. The nextInt() method, which is used to read string.
2. floor(x) method Rounds x to the smallest integer not less than x?
3. To access the elements of an array we use the { } operator.
4. equals method Removes any spaces at the beginning or end of a string.
5. The output of charAt(i) method is int.

Q2 Fill in the blanks (5 marks)

1. The ---------- usually used with the switch-case statement; however, it can be used with a loop statement where it terminates the loop.
2. the string index begins at ----------
3. length Returns the length of the string as ----------
4. int a = (int)18.7; a=----------
5. A ---------- is an indexed list of values that all have the same type.

Q3 What is the output when each of the following statement is executed?(5 marks)

str = " Computer Science";

a) System.out.println(str.trime());

b) System.out.println(str.startsWith ("Java"));

c) System.out.println(Math.abs(7.5));

d) System.out.println(Math.floor(7.5));

e) System.out.println (Math.min(5,6));

Q4: Write a Java program to search an element in an array(10 marks).

Q5: Write down a Java program to reverse the order of elements in an array, for example: A = {12, 0, 3, -4, 7} becomes A= {7, -4, 3, 0, 12}.(5 marks).

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Note: Answer all questions.

Q1 TRUE OR FALSE (5 marks)

1. The nextLine() method, which is used to read int.
2. ceil(x) method Rounds x to the largest integer not greater than x.
1. To access the elements of an array we use the [] operator.
2. toUpperCase method Returns a copy of the original string, all lower case.
3. The output of Substring(i,j) method is int.

Q2 Fill in the blanks (5 marks)

1. In Java, the class --------- provides pre-written mathematical functions to perform various math calculations.
2. The method ---------- used to find the natural logarithm of x (base e).
3. System.out.println(Math.ceil(45.87)); output is----------
4. The method used to compare two strings without considering the case of small or capital letters is ----------
5. The ---------- statement skip the current iteration of a loop and proceeds directly to the next iteration.

Q3 What is the output when each of the following statement is executed?(5 marks)

str = " university of kufa ";

a) System.out.println(str.length());
b) System.out.println(str.startsWith ("Java"));
c) System.out.println(str.Substring(1,4));
d) System.out.println(Math.ceil(7.5));
e) System.out.println (Math.abs(5));

Q4: Write a Java program to sort the items of an array in ascending order(10 marks).

Q5: Write a Java program that performs the following operations on an one-denominational array of type double: Array size [10] (5 marks)

a) Print out odd numbers in the array.

b) Display the values of the third column.

استاذ المادة
م.م هدى عبد الحسن

رئيس القسم
أ.م.د اسعد نوري هاشم

University of Kufa
Faculty of Computer Science & Mathematics
Department: Computer Science
Course Name: Advanced Programming

Academic Year 2020-2021
Course Stage/Level: 1st
Date of Exam: 10/06/2021
Time: One hour

Notes: Attempt all questions

Q1> Answer each of the following statements: (7 marks)

1) What is the output when each of the following statement is executed?

```
String str = "    KUFA UNIVERSITY   ";
System.out.println(str.trime());
System.out.println(startsWith("kufa"));
```

2) The method used to compare two strings without considering the case of small or capital letters is .....

3) The ..... statement skips the current iteration of a loop and proceeds directly to the next iteration.

4) Cast the following float value into integer: int x = 120.5f; .....

5) Correct the error of the following set of code:

```
int count=-11.5;
System.out.println(Floor(Count));
System.out.println(abs(Count));
```

Q2> Write a Java program to find the square root of the smallest number out of five double numbers. (4 marks)

Q3> Write a Java program to read two names and print them in an alphabetical order. (4 marks)

Good Luck